

Quantum Computing

A Quantum Algorithm

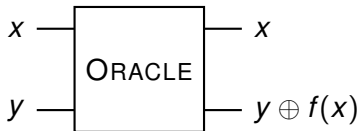


UNIVERSITÀ
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Dipartimento
di **INFORMATICA**

Oracle Computation

- We are given a **black box** containing a computer that is dedicated to computing a particular function $f : \{0, 1\} \rightarrow \{0, 1\}$ using reversible classical computation.
- We can only feed the box with an input at one end and retrieve the output at the other end, after a fixed time that is independent of the input.
- This is called **oracle computation**.



- For the oracle to compute $f(x)$ we must make sure that the auxiliary qubit y is initialised to 0.

Problem 1

Simple task: given an oracle computing $f(x)$, find out what f is.

The only way to do this without looking inside it is to run the oracle once for each possible input.

In		Out	
0	0	0	$f(0)$
0	1	0	$f(0)$
1	0	1	$f(1)$
1	1	1	$f(1)$

Is it possible to perform the task using only one run of the oracle?

No!

General Fact

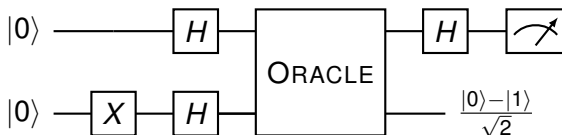
Quantum computation doesn't speed up all computational task but only some of them.

Problem 2

Simple task: given an oracle computing $f(x)$, find out whether f is **constant** or **balanced**.

- How many queries with classical computation?
- How many queries with quantum computation?

Deutsch Algorithm



Answer

The Deutsch circuit solves the problem with certainty after only **one call** to the oracle, whereas any classical circuit would need **two calls** ($n/2 + 1$, in general).

Quantum Interference

Phenomenon that can be explained computationally by the presence of complex amplitudes.

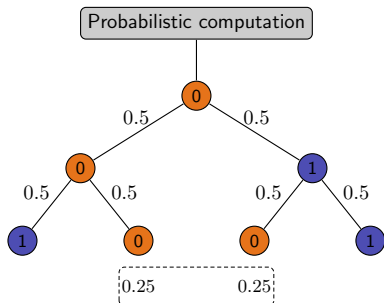


Figure: The probability of '0' is $1/4 + 1/4 = 1/2$

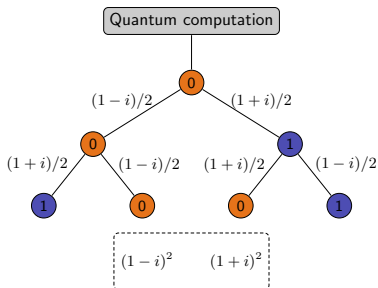
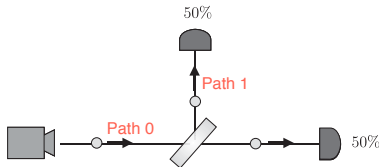


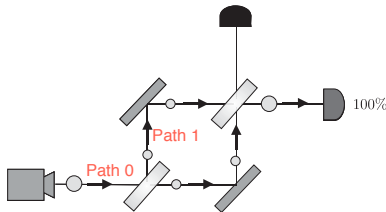
Figure: The probability of '0' is $(1-i)^2 + (1+i)^2 = 0$

Quantum Interference

- Observable phenomenon that cannot be explained by classical physics.
- Phenomenon by which an individual particle, such as a photon (light particles) can cross its own trajectory and interfere with the direction of its path.
- **Experiment:** Measurement statistics with one and two beam splitters

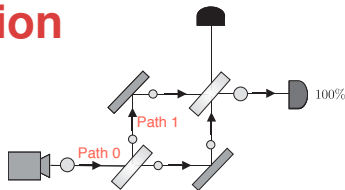


One beam splitter



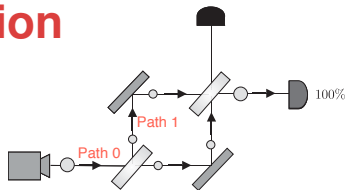
Two beam splitters

Computational Interpretation



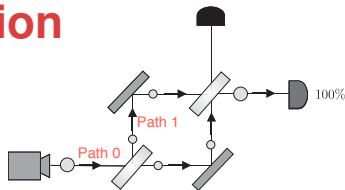
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Computational Interpretation



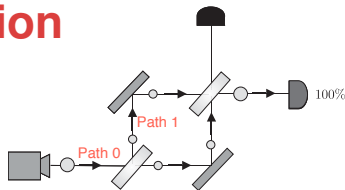
- Looking at the experiment as a computation, we can identify two operations:
 - B for the beam splitter

Computational Interpretation



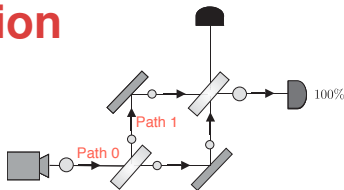
- Looking at the experiment as a computation, we can identify two operations:
 - B for the beam splitter
 - NOT for the full mirror

Computational Interpretation



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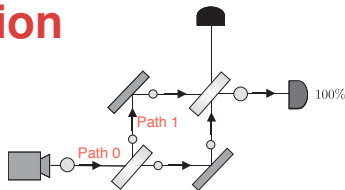
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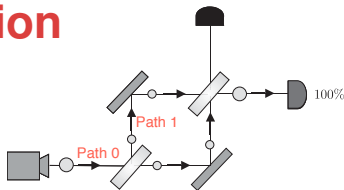


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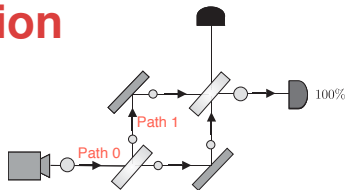


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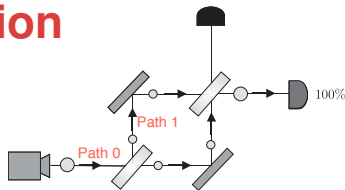


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- Any classical counterpart? NO
- $B^{-1} = \sqrt{X}$

Computational Interpretation



- Looking at the experiment as a computation, we can identify two operations:
 - B for the beam splitter
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- The interference experiment thus becomes:

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- Any classical counterpart? NO
- $B^{-1} = \sqrt{X}$ has no classical analogue among the elementary 1-bit operations.

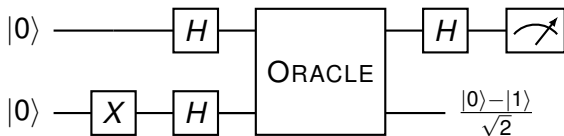
The Deutsch Algorithm

Simple task: given an oracle computing $f(x)$, find out whether f is constant or balanced.

Remark

Contrary to Problem 1, we are asking to determine a property of f , specifically the value of $f(0) \oplus f(1)$.

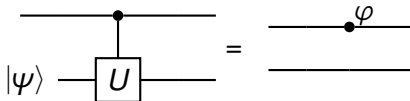
Deutsch Algorithm



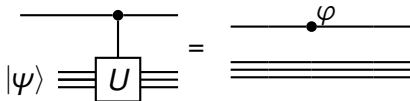
The measurement of the first qubit will give the answer (i.e. the value of $f(0) \oplus f(1)$) with probability 1.

Phase Kick-back

If we apply a controlled- U to a register where the target is prepared in an eigenstate $|\psi\rangle$ of U , then $|\psi\rangle$ stays unchanged and the phase gate $|0\rangle\langle 0| + e^{i\varphi} |1\rangle\langle 1|$ is applied to the control qubit.



This is still the case if the target register contains many qubits.



Phase Kick-back allows us to use a single qubit to read out the action of an arbitrarily large unitary.

Phase Kick-back - Examples

When the target qubit of the $CNOT$ operation is an eigenstate of X , then this eigenstate does not change, and we can effectively treat the eigenvalue as being *kicked back* to the control register.

$$CNOT : |x\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}} \mapsto (-1)^x |x\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Note that $\frac{|0\rangle - |1\rangle}{\sqrt{2}}$ is an eigenstate of NOT with corresponding eigenvalue -1 .

Show that

$$U_f : |x\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}} \mapsto (-1)^{f(x)} |x\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

for $f : \{0, 1\} \rightarrow \{0, 1\}$, and U_f a quantum circuit implementing f .

Hint: U_f can be seen as an operation on the second qubit controlled by the first (input) qubit.

Phase Kick-back

U_f can be seen as the controlled operation $C\hat{U}_f$, with $\hat{U}_f : |b\rangle \mapsto |b \oplus f(x)\rangle$ and $|x\rangle$ the control qubit.



The state $\frac{|0\rangle - |1\rangle}{\sqrt{2}}$ is an eigenstate of $\hat{U}_f$, so it will remain unchanged and the corresponding eigenvalue $(-1)^{f(x)}$ kicked back to the control qubit.

